

USE CASE SPECIFICATION FLOW

Problem Statement #12: Blood Bank Inventory & Emergency Donor Matcher

Document ID: UC-SPEC-01

Target: UC-01 (Core Flow)

Status: Approved Baseline

Use Case ID / Name	UC-01: Submit Emergency Blood Request	Priority / Complexity	High / High
Primary Actor	Emergency Requester (Hospital ER Medic / Authorized Physician)	Secondary Actors	Blood Bank Manager, SMS Gateway, Registered Donors
Relationships	Includes: «include» UC-02: Match Compatible Donors Extension Points: «extend» UC-03: Broadcast Geo-targeted SMS Alert (Triggered upon inventory deficit)		
Preconditions	1. Requester is authenticated and logged into the emergency portal with hospital clearance credentials. 2. Real-time inventory ledger and geolocation services are operational.		
Postconditions	1. Stock Available: Required units are locked/reserved in the ACID ledger, decremented from public pool, and an urgent Dispatch Token is generated. 2. Stock Deficit: Immediate partial reservation occurs, and geo-targeted alerts are dispatched to matching donors within a 10 km radius.		

MAIN SUCCESS SCENARIO (NORMAL ALLOCATION FLOW)

- Request Initiation:** Emergency Requester selects "Create Emergency Request" and inputs blood component (e.g., RBCs, Platelets), ABO/Rh type, required units, urgency flag, and GPS delivery coordinates.
- Input Validation:** System validates request parameters for completeness and cross-checks against active hospital licensing rules.
- Compatibility Check [«include» UC-02]:** System executes UC-02: Match Compatible Donors to check inventory and evaluate universal/compatible donor-recipient matrix tables.
- Stock Verification:** System verifies sufficient unexpired, non-quarantined units exist in the blood bank inventory ledger.
- Atomic Ledger Reservation:** System places an exclusive transactional lock on the matching blood units, changes their state to "Reserved", and decrements available count.
- Dispatch Generation:** System generates an urgent *Dispatch Token & Chain-of-Custody Manifest* with estimated prep time.
- Notification:** System alerts the Blood Bank Manager terminal with pick-and-pack instructions and displays real-time confirmation with the Dispatch Token to the Requester.
- Completion:** Use case terminates successfully.

ALTERNATE & EXCEPTION FLOWS

Alternate Flow 4a: Critical Inventory Deficit – Geo-targeted Donor Broadcast [«extend» UC-03]

1. 4a1. System detects local inventory units are fewer than requested quantity.

2. 4a2. System immediately reserves all available matching units in stock to secure partial fulfillment.

3. 4a3. System extends flow to UC-03: Broadcast Geo-targeted SMS Alert.

4. 4a4. Geospatial engine queries eligible donors within a 10 km perimeter possessing compatible blood types and satisfying the ≥56-day cooldown rule.

5. 4a5. SMS Gateway broadcasts emergency alert payloads with donation center directions to all matched donor devices within 30 seconds.

6. 4a6. Requester UI updates: "Partial stock allocated; emergency broadcast issued to [N] matching donors within 10 km." Flow ends.

Exception Flow 2a: Invalid or Incomplete Request Submission

1. 2a1. System detects missing required fields (e.g., unspecified Rh factor or invalid coordinate format).

2. 2a2. System flags erroneous fields in red and displays specific correction prompts.

3. 2a3. Requester corrects the input data and resubmits. Flow resumes at Step 2 of the Main Success Scenario.